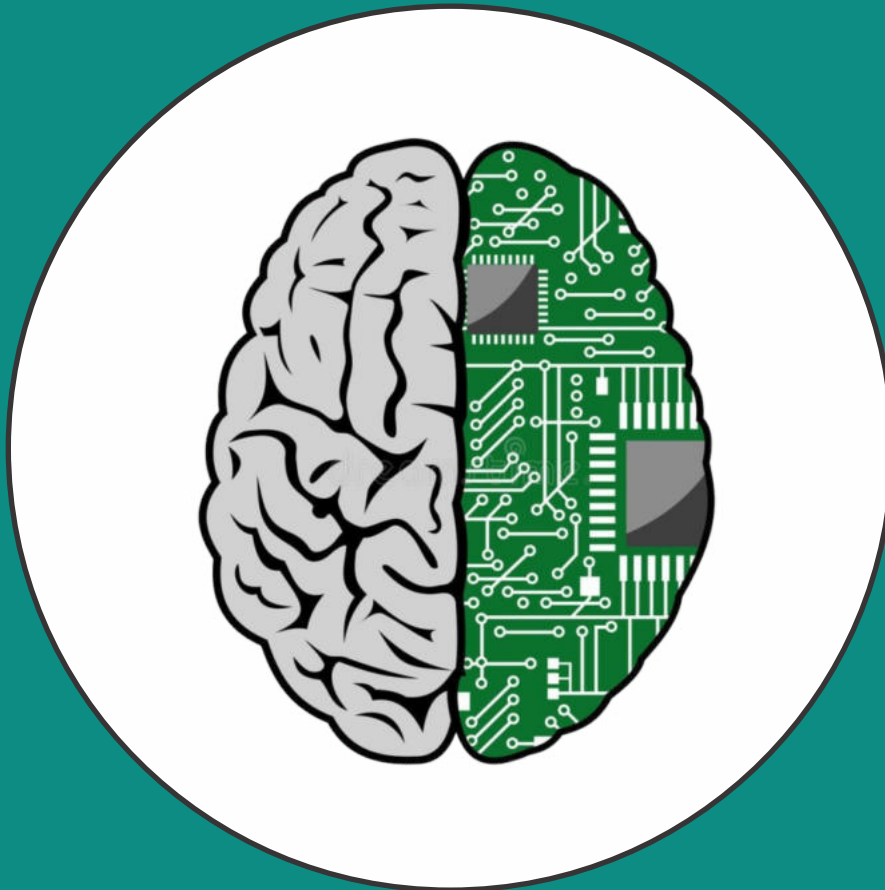




UNIT 5

COMPUTER MEMORY



Learning Objectives

- Understand computer memory as storage for data and information.
- Distinguish between primary (RAM and ROM) and secondary memory.
- Describe primary memory as temporary (RAM) or permanent (ROM).
- Recognize secondary memory devices (e.g., hard disk, USB) and their capacities.



COMPUTER MEMORY

UNIT 5

COMPUTER MEMORY

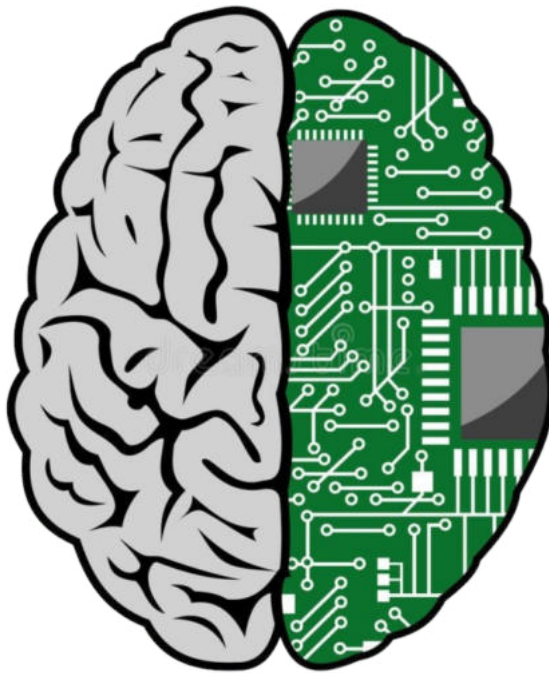
Computer memory is where data and information are stored, either temporarily or permanently.

- **DATA**

Data is a collection of facts and figures, like words, numbers, and symbols.

- **INFORMATION**

Information is data that has been processed so we can understand it better.



There are two main types of computer memory:

1. Primary Memory
2. Secondary Memory



PRIMARY MEMORY



UNIT 5

COMPUTER MEMORY

Primary memory is the main memory of the computer. The computer needs this memory to work. It stores only the data and instructions the computer is using at the moment. It has **limited** space.

There are two types of primary memory:

1. RAM (Random Access Memory)
2. ROM (Read Only Memory)

- **RAM (RANDOM ACCESS MEMORY)**

- RAM is a temporary memory. When the computer starts, important programs load into RAM.
- The data in RAM is lost when the computer is turned off, so it's called **volatile** or **temporary memory**.

- **ROM (READ ONLY MEMORY)**

- ROM is permanent memory. It stores basic instructions for starting the computer.
- The data in ROM is not lost when the computer is turned off, so it's called **non-volatile** or **permanent memory**.



SECONDARY MEMORY

Secondary memory stores data and information permanently. It has much more space than primary memory and is called **non-volatile memory** or **secondary storage**.



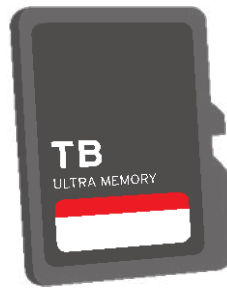
UNIT 5

COMPUTER MEMORY

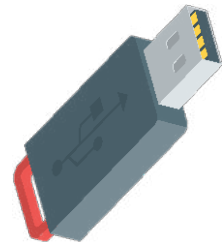
Some devices used as secondary memory are:



Hard Disk



Memory Card



USB Flash Drive



DVD



CD

• HARD DISK

- The hard disk is the most common secondary storage device, located inside the system unit.
- Hard disks can store large amounts of data, such as 256 GB, 500 GB, 1 TB, and more.

• USB FLASH DRIVE

- A USB flash drive is a portable storage device, also called a data traveller or pen drive.
- It plugs into the computer's USB port, and we can read and write data on it. It is often used to transfer data between computers.

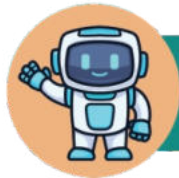
- **CD/DVD AND BD**

- A CD (compact disk), a DVD (digital video disk) and BD (blu-ray disc) store data like pictures, movies, and music.
- CDs can store about 700 MB, DVDs can store up to 8.5 GB, and Blu-Ray discs can store up to 25-128 GB.



UNIT 5

COMPUTER MEMORY



EXERCISE 1

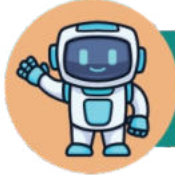
Fill in the blanks.

1. Primary memory is the _____ memory of the computer.
2. RAM is a _____ memory.
3. The hard disk is located _____ the system unit.
4. A CD can store data up to _____.
5. ROM stands for _____.
6. _____ is often used to transfer data between computers.



UNIT 5

COMPUTER MEMORY



EXERCISE 2

Mark "True" or "False".

TRUE

FALSE

CD stands for Compact Disk.

☐☐

A USB flash drive is a portable device.

☐☐

When the computer starts, programs load into ROM.

☐☐

A USB flash drive is also known as a data traveller.

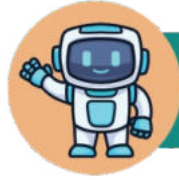
☐☐

RAM is non-volatile memory.

☐☐

Hard disk is the main memory of the computer.

☐☐



EXERCISE 3



UNIT 5

COMPUTER MEMORY

Answer the following questions.

1. Differentiate between “data” and “information”.

Ans: _____

2. What is computer memory?

Ans: _____

3. What is primary memory?

Ans: _____

4. What is secondary memory?

Ans: _____

5. How much data can a hard disk store?

Ans: _____

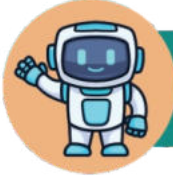
6. How much data can a CD and a DVD store?

Ans: _____



UNIT 5

COMPUTER MEMORY



EXERCISE 4

Match the columns:

Column A

Data is a collection of

Information is

The computer cannot work without

USB flash drive consists of

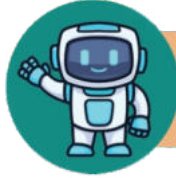
Column B

memory chips

primary memory

processed data

facts and figures

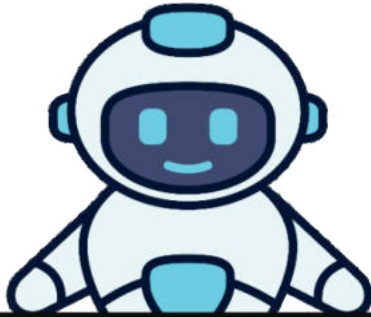


LAB ACTIVITY



UNIT 5

COMPUTER MEMORY



1. Go to the computer lab, turn on the computer, and open MS Paint.
2. Draw a simple picture.
3. Save the drawing to a **USB Flash Drive** by clicking "File," then "Save As," selecting the USB Drive, and giving your file a name.

TEACHER'S NOTE

Help students understand the differences between primary and secondary memory, and provide examples of each type.